

A nearly quantitative synthetic approach towards monocyclic polystyrenes and the solvent, concentration and molecular weight effect on cyclic yield

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ABSTRACT

We report herein the effects of solvent, concentration, and molecular weight of α -alkynyl- ω -azido polystyrenes (*l*-PS- N_3) on the synthesis of monocyclic polystyrenes (*c*-PS) by intra-chain coupling of *l*-PS- N_3 via Cu-catalyzed azide/alkyne cycloaddition method. By using fully end-functionalized *l*-PS- N_3 as precursors, the optimum cyclization conditions to synthesize high purity *c*-PS (>99% by SEC) are revealed, at concentrations around 7.9 μ mol/L in DMF. The *c*-PS products are fully characterized by SEC, LCCC, ¹H NMR, and MALDI-TOF Mass spectrometry. The yield of *c*-PS decreases with increment of concentration of *l*-PS- N_3 as predicted by Jacobson and Stockmayer's equation. Interestingly, among the three studied PS samples with different molecular weight (~4k, 9k, 14k Da), the medium one (PS9k) provides the best *c*-PS yield, while in the three studied solvent (toluene, DMF, cyclohexane), DMF gives the best yield. The results reveal that the PS chain conformation in the solution, which is affected by the solvent and molecular weights, plays an important role in the intra-chain cyclization reaction.

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1. Introduction

Due to the unique cyclic topological structure and the absence of “chain ends”, cyclic polymers show some interesting properties compared to their linear counterparts, including higher refractive index and mass density [1–3], smaller hydrodynamic volume and radius of gyration [4,5], reduced viscosity [6–9], and increased rate of crystallization [10,11]. The ring-expansion polymerization [12–17] and ring-closure methods [18–26] are the two main strategies to prepare cyclic polymers. The big obstacle among these strategies is the contamination of linear counterparts in the cyclics, which hinders the mass production of cyclic polymers since a tedious purification process is needed in order to remove the linear byproducts, as it is well-known that the cyclic polymers' properties are affected greatly by the contamination of a small portion of

linear counterparts [27–31].

Recently, Laurent and Grayson [21] developed a facile method to synthesize cyclic polymers, where living/controlled polymerization was used to synthesize the linear polymers, followed with the Cu-catalyzed azide/alkyne cycloaddition (CuAAC) [32,33] intra-chain “click” reaction to synthesize monocyclic polymers after the chain-end transformation of the corresponding linear precursors. The strategy is widely accepted and a variety of cyclic polymers have been synthesized [34–40]. However, few publications on click cyclization have focused on a detailed quantification of cyclic purity, but that purity is a critical issue for the future of this field [29].

Alternatively, we envisage that the quantitative synthesis of well-defined cyclic polymers would help us avoid the contamination of linear polymers and the purification process, and open out the way to the mass production of cyclic polymers. In principle, there are three major questions hindering quantitative synthesis of cyclic polymers: 1) Can we achieve fully α,ω -functionalized linear polymer precursors? 2) Can we achieve high coupling reaction efficiency (~100%) between the functional end-groups? 3) Can we

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avoid the inter-chain coupling side reactions between α,ω -functionalized linear polymers? Despite extensive studies on the synthesis of cyclic polymers, the above three questions have not been sufficiently addressed, and the quantitative synthesis of cyclic polymer has not been reported.

In this work, we report our approach towards the quantitative synthesis of monocyclic polymers based on the atom transfer radical polymerization (ATRP) [41–44] and CuAAC reaction. The synthetic route is presented in Scheme 1. Styrene is selected as monomer since its ATRP process is easy to be controlled and polystyrenes (PS) are easy to characterize. All the three aforementioned questions are addressed. Namely, to achieve fully α,ω -functionalized linear polymer precursors, the conversions of linear PS (*l*-PS) synthesis by ATRP are controlled to be less than 20% to minimize the termination process [45]; the ring-closure “click” reaction was utilized to achieve high coupling efficiency; and the reaction was carried out in dilute conditions to suppress the inter-chain coupling reaction. The effects of solvents, concentration and molecular weights of *l*-PS on the monocyclic PS yield were studied, and the optimum conditions for the synthesis of monocyclic PS was revealed.

2. Experimental section

2.1. Materials

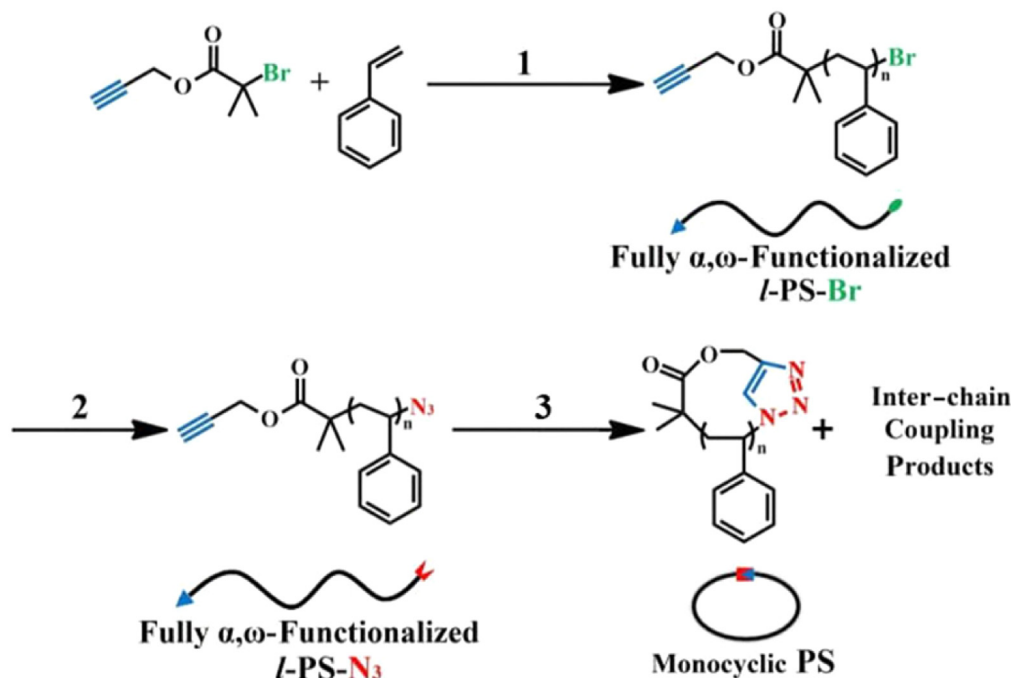
All chemicals were obtained from Sinopharm Group Chemical Reagent Co., Ltd. (China) unless mentioned below. Prior to use, *p*-xylene (AR, 98.5%) was washed three times with concentrated sulfuric acid (AR, 98%) and twice with aqueous 5% NaHCO_3 (AR, 99.8%) and distilled deionized water, dried with anhydrous MgSO_4 (AR, 98%), and then distilled from CaH_2 (AR, 98%). Styrene (AR, 99%) was washed with 10% NaOH (AR, 99%) aqueous solution and deionized water to remove inhibitors, dried with anhydrous MgSO_4 for several hours, then distilled under vacuum and stored in refrigerator. Copper (I) bromide (CuBr , CP, 99%) was washed twice by glacial acetic acid (AR, 98%) and acetone (AR, 99.8%) respectively

and stored under nitrogen. Tetrahydrofuran (THF, AR, 99.5%), dichloromethane (AR, 99.5%) and triethylamine (TEA, AR, 99%) were distilled from CaH_2 under nitrogen immediately before use. Neutral alumina (pH 6.5–7.5, 100–200 mesh), chloroform (AR, 99.8%), and methanol (AR, 99.8%) were used as received. *N,N*-dimethylformamide (DMF, AR, 99.5%) was purchased from Shanghai Chemical Reagents Co. (Shanghai, China) and dried with activated molecular sieves (4 Å). *N,N,N',N'',N'''*-pentamethyldiethylenetriamine (PMDETA, 99%) and sodium azide (NaN_3 , 99.5%) were purchased from Sigma-Aldrich and used as received. Propargyl 2-bromoisobutyrate was synthesized according to the method by Han et al. (Scheme S1) [38].

Synthesis of *l*-PS-Br. A slightly modified ATRP procedure was applied here: a dry Schlenk tube was charged with propargyl 2-bromoisobutyrate, styrene, PMDETA, CuBr and *p*-xylene. The reaction mixture was degassed by three standard freeze-pump-thaw cycles, and placed in an oil bath thermostated at 85 °C polymerized for a setting time. After that, the solution was cooled, diluted with THF and passed through a column of neutral alumina to remove the metal salt. After precipitation into a large excess of methanol, the white polymer was collected by filtration, dried at 40 °C under vacuum overnight and characterized by size exclusion chromatography (SEC). Detailed recipe for the synthesis of *l*-PS-Br with different molecular weights is provided in Table S1.

Synthesis of *l*-PS- N_3 . A typical synthetic procedure is described as following: into a 100 mL round-bottom flask with a magnetic stirrer, *l*-PS-Br (0.25 mmol), DMF (50 mL) and NaN_3 (2.5 mmol) were added. After the reaction mixture was stirred at room temperature for 24 h, PS was precipitated into a large excess of methanol, collected by filtration, and washed with excess water and methanol. The white powder was dried at 40 °C under vacuum overnight.

Synthesis of Cyclic PS (*c*-PS). The synthetic procedure for the preparation of *c*-PS-4k in DMF at 7.9 $\mu\text{mol/L}$ (final concentration) is given below as an example. A DMF (300 mL) solution of Cu(I)Br (20 mg, 0.14 mmol) and PMDETA (49 mg, 0.28 mmol) was purged with argon for 3 h at 60 °C, after which a DMF (400 mL) solution of



Scheme 1. Synthetic route for cyclic polystyrenes: 1) CuBr , PMDETA, *p*-xylene, 85 °C; 2) NaN_3 , DMF, 24 h, room temperature; 3) CuBr , PMDETA, 60 °C.

l-PS4k-N₃ (20 mg, 0.0050 mmol) was added dropwise to the solution with the dropping rate of 0.55 mL/min. The reaction mixture was stirred further for 12 h to ensure complete cyclization after the addition of the linear precursor solution. The solvent was removed under vacuum and the residue was diluted with chloroform. The copper salts were removed by passing the reaction mixture through a neutral alumina column and the chloroform was removed by rotation evaporation. After precipitation into methanol, the cyclic polymer was collected after filtration and drying under vacuum for 24 h at 40 °C. For the concentration effect study, different amounts of *l*-PS-N₃ were dissolved in 400 mL solvent and dropped into 300 mL solvent containing catalyst, with the final PS concentration of 7.9 μmol/L, 15.7 μmol/L and 31.4 μmol/L, while PS with molecular weights of 4k, 9k and 14k Da were used to study the effect of molecular weights. Toluene or cyclohexane was used instead of DMF for solvent effect study. Other conditions including the solution dropping rate and stirring rate were similar for all the reactions. For clarification, the products obtained with suffix “1”, “2” and “3” after the sample name denote the final PS concentration of 7.9 μmol/L, 15.7 μmol/L and 31.4 μmol/L, respectively, while “D”, “T” and “C” denote the solvent of DMF, toluene and cyclohexane, respectively.

2.2. Characterization

¹H nuclear magnetic resonance (NMR) spectra were recorded on Agilent Technologies 600 MHz DD2 spectrometer with PFG ¹H/¹⁹F/X probe at 25 °C, using deuterated dichloromethane as the solvent and tetramethylsilane (TMS) as the internal reference. For ¹H NMR measurements, an improved quantitative method were used here following our previous reported work, where the decoupling of ¹³C to ¹H was applied during acquisition, with the delay time (20s) set as 10 times the relaxation time (T₁), and the experiments were conducted at 90° pulse width for the maximum signal acquisition [46,47].

Size Exclusion Chromatography (SEC) analysis was performed on a modular system comprising a Waters 1515 pump, a 717 plus autosampler, and a 2410 refractive index detector. Separations were achieved using three PL Mixed-C (7.5 × 300 mm, 5 μm bead size, 200–2000000 Da) columns, calibrated with 9 narrow distribution polystyrene standards with a molecular weight range from 474 to 177000 Da. All samples (2 mg/mL) were passed through 0.45 μm PTFE filter before analysis and THF was used as mobile phase at a flow rate of 1.0 mL/min at 35 °C.

Liquid chromatography at critical condition (LCCC) was conducted with a single C18 bonded silica column (Nucleosil bare silica, 50 Å pore, 250 × 4.6 mm, 5 μm bead size), following the procedures as reported [29,48–50]. The mobile phase was a mixture of THF/*n*-hexane (42/58, v/v), and the flow rate was 0.5 mL/min at 43 °C.

Fourier transform infrared (FTIR) spectra were collected on Nicolet 6700 FT-IR Instrument. Samples were dispersed in potassium bromide uniformly and compressed into pellets. The spectra were recorded with a resolution of 4 cm^{−1} co-adding 32 scans with the range from 400 cm^{−1} to 4000 cm^{−1}.

Matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrum was acquired on a Bruker Ultraflex-Treme TOF/TOF mass spectrometer (Bruker Daltonics, Inc., Billerica, MA) equipped with a Nd:YAG laser (355 nm). All spectra were recorded in linear mode. *Trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB, Aldrich, >99%) was used as matrix at a concentration of 20 mg/mL in CHCl₃ while sodium trifluoroacetate (CF₃COONa) as the cationization agents which were prepared in ethanol at a concentration of 10 mg/mL. The matrix and cationization agents were mixed with the ratio of

10/1 (v/v). The sample preparation involved depositing 0.5 μL of matrix and salt mixture on the wells of a 384-well ground-steel plate, depositing 0.5 μL of the sample (10 mg/mL in CHCl₃) on a spot of dry matrix, and adding another 0.5 μL of matrix and salt mixture on top of the dry sample. After evaporation of the solvent, the plate was inserted into the MALDI source. The mass scale was calibrated externally using the peaks obtained from protein standard (4000–20000 Da), and the data were analyzed by Bruker's flexAnalysis and PolyTool software.

3. Results and discussion

Controlled synthesis of fully α,ω -functionalized linear polymers is the key for the synthesis of cyclic polymers. Lutz et al. reported highly end-functionalized polymers can be obtained by stopping the ATRP at low conversion [51,52]. Following their strategy, three α -alkynyl- ω -bromopolystyrenes (*l*-PS-Br) with different molecular weights were synthesized at low yield (<20%), and used as model compounds to study the influence of molecular weights on cyclization, with their SEC curves presented in Fig. 1. All the curves have a narrow and symmetric distribution shape with the calculated number-average molecular weights relative to PS standards of 3950 Da, 9090 Da and 14200 Da, and corresponding polydispersity index of 1.10, 1.07 and 1.07, respectively. The symmetric SEC curves and narrow molecular weights distribution suggest the well-controlled characteristic during polymerization.

Quantitative NMR experiments [46,47] of *l*-PS-Br were conducted to estimate the end-functionality of polystyrenes with bromine and alkynyl groups. The spectra of *l*-PS4k-Br are presented in Fig. 2A, with the assignment of each peak corresponding to its chemical structure, while *l*-PS9k-Br in Fig. S1A and *l*-PS14k-Br in Fig. S2A, respectively. In Fig. 2A, the ratio of protons' peak integration area from the methylene group adjacent to the alkynyl group (*H_b*, at α chain end position) to that of the methine group connected with bromine and phenyl groups (*H_g*, at ω chain end position) is 2.00:0.97, similar to the theoretical value of 2:1, indicating a high extent of end group fidelity, within the margin of error of the NMR technique. Similar results were observed for *l*-PS9k-Br and *l*-PS14k-Br. In addition, by comparing the ratio of peak area of *H_b* to that of the phenyl groups (*H_h*), the molecular weights of PS can be obtained, which is 4070 Da, 9790 Da and 14900 Da for *l*-PS4k-Br, *l*-PS9k-Br, and *l*-PS14k-Br, respectively, similar to the

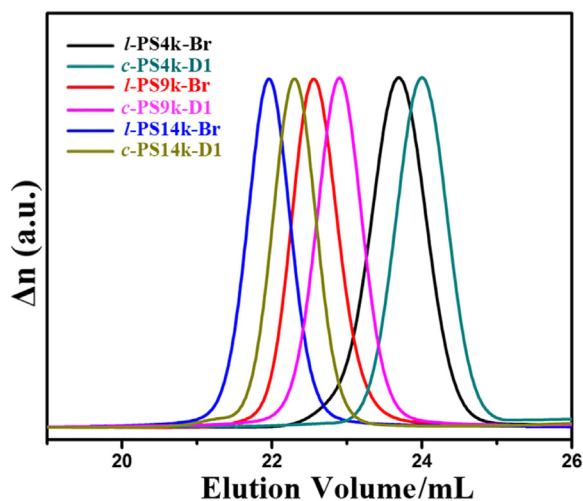


Fig. 1. SEC curves of linear polystyrenes (*l*-PS-Br) and cyclic polystyrenes (*c*-PS-D1) with different molecular weights.

molecular weights obtained from SEC. The accordance of the molecular weights from SEC and NMR validates our data, and further supports the fully end-functionality of *l*-PS-Br.

The substitution of bromine at PS chain end by azide group is considered to be quantitative [53,54]. The SEC curves of *l*-PS- N_3 are superimposed with *l*-PS-Br precursors and provided in Fig. S3, and the end-functionality of *l*-PS- N_3 was determined by quantitative 1H NMR, where the spectrum of *l*-PS4k- N_3 is shown in Fig. 2B, while *l*-PS9k- N_3 in Fig. S1B and *l*-PS14k- N_3 in Fig. S2B, respectively. After reacted with sodium azide, the peaks from protons at ω end adjacent to bromine (H_g , 4.5 ppm) in *l*-PS-Br were vanished, indicating the full conversion of bromide chain ends. The corresponding new peak for H_{g1} overlapped with H_{b1} after substitution, similar to work reported by Tsarevsky et al. [44]. The integration ratio from peaks at 4.0 ppm ($H_{g1} + H_{b1}$) to that from the phenyl groups (H_h) is 3.00:193 for *l*-PS4k- N_3 , by which the calculated molecular weight is 4180 Da, similar to the value from *l*-PS4k-Br (4070 Da), suggesting the fully conversion of bromine to azido group. Similar results were observed in the spectra of *l*-PS9k- N_3 and *l*-PS14k- N_3 (Table S2). These fully end-functionalized α -alkynyl- ω -azido polystyrenes (*l*-PS- N_3) were used to synthesize *c*-PS in DMF by a CuAAC ring-closure method, and the influence of concentration on cyclization was studied.

During the cyclization process, there always exists the competition between intramolecular cyclization and intermolecular coupling reaction. Therefore, the cyclization reaction of α -alkynyl- ω -azido polymer should be carried in high dilute conditions to suppress the intermolecular reaction [21,55]. For an AB ring closure system, the probability of forming intra-chain monocyclic polymers (P_C) and inter-chain coupling products (P_L) follows the Jacobson and Stockmayer's equation when the chains can be regarded as random coils [56]:

$$P_C = \left(\frac{3}{2\pi\langle r^2 \rangle} \right)^{\frac{3}{2}} v_s \quad (1)$$

$$P_L = \frac{2N_A C}{M} v_s = 2N_A [C_M] v_s \quad (2)$$

where $\langle r^2 \rangle$ is the mean-square end-to-end distance of the polymer chain, v_s the small volume of chain ends within which the reaction carries (also called the capture volume by Monteiro et al.) [17,33], N_A the Avogadro's number, M the molecular weight of polymer, C the concentration of polymer, and $[C_M]$ the molar concentration of polymer (in mol/m³). The yield of monocyclic polymers (w_c) can thus be obtained:

$$w_c = \frac{P_C}{P_C + P_L} = \frac{1}{1 + 2^{\frac{3}{2}} N_A [C_M] \left(\frac{\pi \langle r^2 \rangle}{3} \right)^{\frac{3}{2}}} \quad (3)$$

The end-to-end distance of PS random coil can be related to their molecular weights (M) by the following equation [57]:

$$\langle r^2 \rangle / nm^2 = (7.0 \times 10^{-2})^2 M \quad (4)$$

Since the molecular weights of PS synthesized by ATRP are around 10000 Da, by substitution Eq. (4) into Eq. (3), and considering $w_c > 99\%$ as the quantitative yield, for a PS system with molecular weights of 10000 Da, the following equation can be deduced:

$$[PS] < 8.0 \times 10^{-3} \text{ mol/m}^3 = 8.0 \mu\text{mol/L} \quad (5)$$

The CuAAC reaction of *l*-PS- N_3 with different molecular weights was carried in DMF solution first with concentration of 7.9 $\mu\text{mol/L}$, which is selected to be slightly lower than the theoretical value to achieve the quantitative yields. The SEC curves of the corresponding products (*c*-PS-D1) are shown in Fig. 1. Clearly, the elution volume increased after reaction, indicating the formation of cyclic polymers since they have smaller hydrodynamic volumes than the corresponding linear precursors. For *c*-PS4k-D1 and *c*-PS9k-D1, only a smooth symmetric SEC curve was observed without any shoulder peak, suggesting the nearly quantitative formation of monocyclic PS as no inter-chain coupling products was observed. For *c*-PS14k-D1, a tiny shoulder peak was observed at the left side of main peak, which is about 1% of the main peak area calculated by Origin's Fit Multi-Peaks software analysis. This is from the inter-chain coupling products, which is reasonable based on Eq. (3) considering the larger molecular weight of *l*-PS14k- N_3 . The problem can be solved by carrying out the CuAAC reaction at lower concentrations.

Fig. S4A is the SEC curve of the product from the cyclization of *l*-PS14k- N_3 at concentration of 6.0 $\mu\text{mol/L}$ in DMF. Only the curve for monocyclic PS is observed while the shoulder peak is suppressed, and the narrow unimodal curve is superimposed with the main curve of *c*-PS14k-D1, indicating the nearly quantitative yield of monocyclic PS. For comparison, similar reaction conditions were used for the cyclization of *l*-PS9k- N_3 , and the product's SEC curve is shown in Fig. S4B. The narrow unimodal curve is superimposed with *c*-PS9k-D1, confirming the nearly quantitative synthesis of monocyclic PS when the cyclization reaction of *l*-PS- N_3 was carried at lower concentrations.

It should be noted that these cyclic products, like previous reported materials using the click cyclization, were characterized without further purification, i.e., no azide resin treatment or SEC fractionation or HPLC fractionation of the obtained product was carried after the synthesis of cyclic polymers [48,58–60]. Thus when a nearly quantitative yield is achieved, no purification process is needed. However, the “quantitative” yield here doesn't mean the absolutely 100% conversion of linear precursors to monocyclic polymers, but near 100% conversion (>99%) that the products from side reactions (inter-chain coupling) can't be detected by SEC characterization.

The high conversion of *l*-PS- N_3 to monocyclic products is confirmed by the quantitative NMR spectra, where *c*-PS4k-D1 is presented in Fig. 2C, while *c*-PS9k-D1 in Fig. S1C and *c*-PS14k-D1 in Fig. S2C. Compared to *l*-PS- N_3 , the peaks at 4.0 ppm corresponding to methylene group adjacent to alkynyl group (H_{b1} , at α chain end position) and methine group connected with azide and phenyl group (H_{g1} , at ω chain end position) disappeared, while two peaks at 4.5 ppm and 5.0 ppm assigned to the methylene (H_{b2}) and methine groups (H_{g2}) connected with the triazole ring appeared. For *c*-PS4k-1, the peak integration ratio of H_{b2} to H_{g2} is 2.00:0.97, in agreement with the theoretical value (2:1), indicating the complete transformation of alkynyl and azide groups to triazole groups. This is also supported by the IR spectra of *l*-PS4k- N_3 and *c*-PS4k-1 (Fig. S5), where the bands at 2110 cm⁻¹ and 3295 cm⁻¹ corresponding to the absorption of azide and alkynyl groups disappeared completely after cyclization. In addition, the peak integration ratio of H_{b2} to phenyl groups (H_h) is 2.00:188, from which the calculated molecular weight is 4080 Da, in accordance with the value for *l*-PS4k- N_3 (4180 Da). Similar results were observed in the spectra of *c*-PS9k-D1 and *c*-PS14k-D1 (Table S2). Combined with the SEC results, the similar molecular weights but larger retention volumes of *c*-PS-D1 to their *l*-PS- N_3 precursors clearly indicate the nearly quantitative formation of monocyclic polymers.

The formation of monocyclic PS was further confirmed by the MALDI-TOF mass spectra of cyclic PS and their linear precursors, as

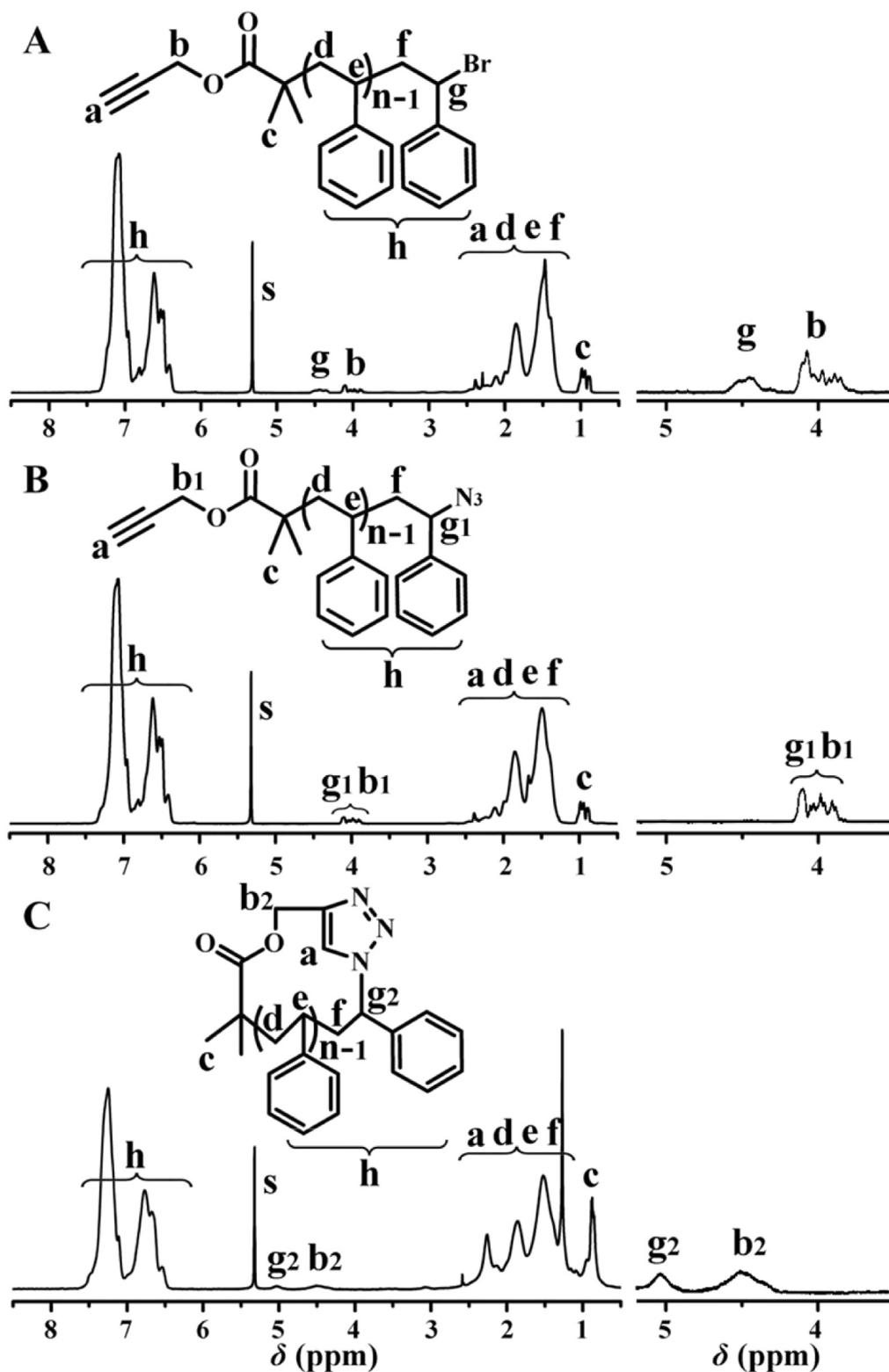


Fig. 2. Quantitative full and expanded ^1H NMR spectra of linear polystyrene precursors (A: *l*-PS4k-Br, and B: *l*-PS4k- N_3) and cyclic polystyrenes (C: *c*-PS4k-D1) in CD_2Cl_2 . s: solvent.

presented in Fig. 3A for *l*-PS4k- N_3 and Fig. 3B for *c*-PS4k-D1, respectively, with the central peaks enlarged at the right side. All spectra were recorded in linear mode. In Fig. 3A, there are a series of major peaks with the shape of narrow and symmetric distribution. The difference of peaks to their adjacent peaks in the same

series is about 104.1 (m/z), exactly the value for styrene monomer, with the peaks assigned to $[\text{l-PS-}\text{N}_3+\text{Na}]^+$. For example, the peak at 5293.23 (m/z) among the main peak series is assigned to $[\text{l-PS}_{49}\text{-}\text{N}_3+\text{Na}]^+$ ($\text{C}_{399}\text{H}_{401}\text{N}_3\text{O}_2\text{Na}^+$, theoretical value: 5293.46). There are also a series of small peaks observed at the left side of major peaks

with mass over charge (m/z) difference of about 23.3, which is ascribed to the metastable peaks as reported by Graysons et al. [21,61,62]. Besides the two series of peaks, no other significant peak was observed.

As for *c*-PS4k-D1 in Fig. 3B, the main peak series are similar to that from *l*-PS4k-N₃ with nearly the identical values, which can be ascribed to $[c\text{-PS} + \text{Na}]^+$, while the other small peaks to metastable peaks which may come from the linear precursor contaminations [61,62]. The molecular weights calculated from the main peaks by MALDI-TOF mass spectra are similar for cyclic polymers and their linear precursors, with 4980 Da for *c*-PS4k-D1 and 5050 Da for *l*-PS4k-N₃, respectively. Similar results were observed for *c*-PS9k-1 (8770 Da) and *l*-PS9k-N₃ (8670 Da) from Fig. S6, and *c*-PS14k-1 (13800 Da) and *l*-PS14k-N₃ (14100 Da) from Fig. S7, respectively. Combined with the SEC results, the fact that *c*-PS-D1 and corresponding linear *l*-PS-N₃ precursors have similar molecular weights (by MALDI-TOF Mass) but different elution volume (by SEC) again proves the formation of cyclic polymers by the monocyclic addition from linear precursors.

As pointed above, the synthesis of fully α,ω -functionalized linear polymers is one of the concerns in the quantitative synthesis of cyclic polymers, and was achieved by carrying the ATRP in low yield (<20%) in our experiments. To prove this point, a control experiment was carried out, where the PS sample (*l*-PS-Br-ct) was synthesized at 36% yield with molecular weights of 6730 Da and PDI of 1.11. The same reaction procedure and condition for the nearly quantitative synthesis of *c*-PS-D1 was applied. Fig. S8 is the SEC curves for this sample before and after cyclization. After cyclization, there is a small shoulder peak remaining at the left side of main peak, which can be ascribed to the linear PS with “dead” ends. This control experiment proves the importance in controlling the monomer conversion to achieve highly α,ω -functionalized linear polymers, and opens out a feasible ATRP process to them.

Concentration effect: As the solution volume and the dropping rate is same for the synthesis of all samples, while the real-time cyclization concentration of *l*-PS-N₃ is smaller than the final PS concentration but hard to be detected, the final concentration is

used instead for the study of concentration effect. Three concentrations of 7.9 $\mu\text{mol/L}$, 15.7 $\mu\text{mol/L}$ and 31.4 $\mu\text{mol/L}$ were chosen to carry the CuAAC reaction, where the first one for the study of quantitative synthesis as discussed above, while others for the study of concentration effect. Fig. 4 is the SEC curves of the products from the cyclization of *l*-PS4k-N₃ at different concentrations. At concentrations of 15.7 $\mu\text{mol/L}$ (*c*-PS4k-D2) and 31.4 $\mu\text{mol/L}$ (*c*-PS4k-D3), a small shoulder peak was observed at the left side of the main peak. Each curve can be separated into two peaks, with the major one representing the monocyclic PS which overlaps with the curve of *c*-PS-D1, while the small peak from the inter-chain coupling products of *l*-PS4k-N₃. The calculated area percentage of the main peak is 98% for *c*-PS4k-D2 and 96% for *c*-PS4k-D3, respectively, which is the yield of monocyclic PS. As predicted, a higher linear precursor concentration leads to lower monocyclic polymer yield. The results are summarized in Table 1.

Molecular weight effect: The cyclization of *l*-PS9k-N₃ and *l*-PS14k-N₃ were conducted in DMF using the similar reaction conditions as *l*-PS4k-N₃ to study the molecular weight effect. The SEC curves of the resultant polymers from different cyclization concentrations are presented in Fig. S9 (*c*-PS9k) and S10 (*c*-PS14k). Similar to *c*-PS4k, at concentrations of 15.7 $\mu\text{mol/L}$ and 31.4 $\mu\text{mol/L}$, a small shoulder was observed at the high molecular weights side of main curve, which is ascribed to the inter-chain coupling products, and the yield of monocyclic PS decreased with increasing concentration of PS precursors. At the same concentrations, the yield of monocyclic PS from *l*-PS9k-N₃ is similar to that from *l*-PS4k-N₃, while that from *l*-PS14k-N₃ is slightly smaller than the other two. This is reasonable according to Eq. (3), where a higher molecular weight PS sample should lead to lower monocyclic PS yield.

Solvent effect: The cyclization of linear PS precursors was also carried in toluene and cyclohexane to study the influence of solvent on the synthesis of cyclic polymers. Fig. 5 is the SEC curves of products after the cyclization of *l*-PS4k-N₃ in toluene at concentrations of 7.9 $\mu\text{mol/L}$, 15.7 $\mu\text{mol/L}$ and 31.4 $\mu\text{mol/L}$, while *l*-PS9k-N₃ in Fig. S11 and *l*-PS14k-N₃ in Fig. S12, respectively. The yields of the monocyclic PS were calculated and the results are listed in Table 1.

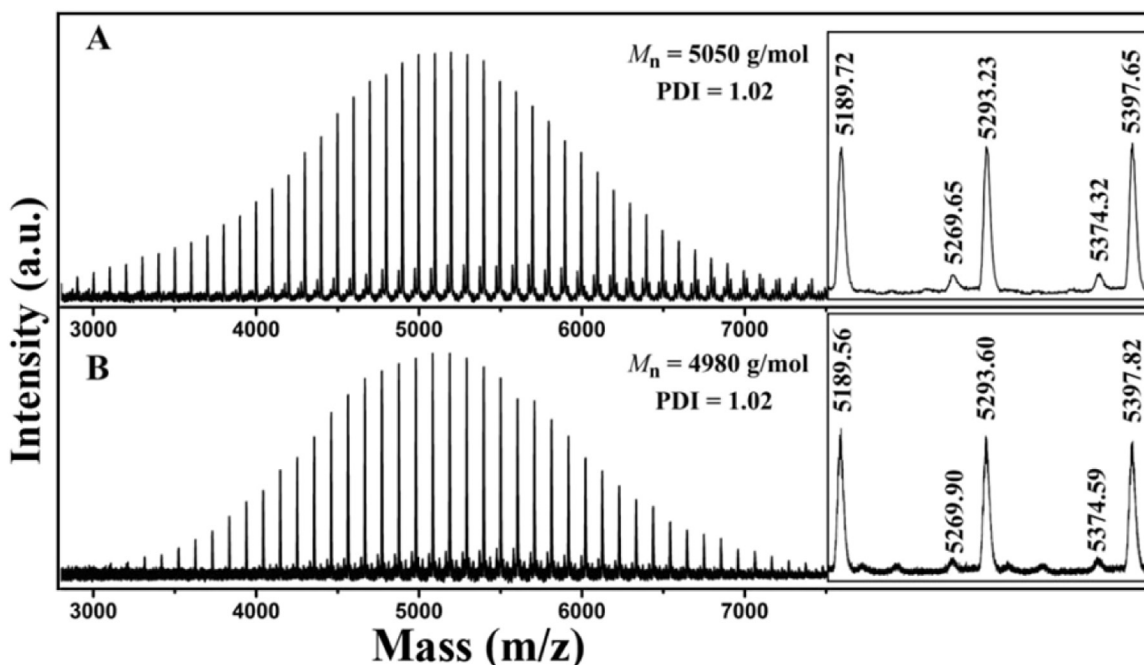


Fig. 3. MALDI-TOF mass spectra of linear polystyrene precursor (A: *l*-PS4k-N₃) and cyclic polystyrene (B: *c*-PS4k-D1).

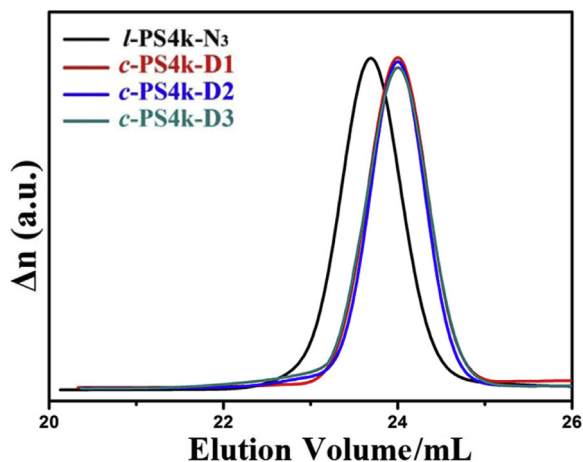


Fig. 4. SEC curves of *l*-PS4k- N_3 precursor and the corresponding cyclization product in DMF at concentrations of 7.9 $\mu\text{mol/L}$ (*c*-PS4k-D1), 15.7 $\mu\text{mol/L}$ (*c*-PS4k-D2), and 31.4 $\mu\text{mol/L}$ (*c*-PS4k-D3).

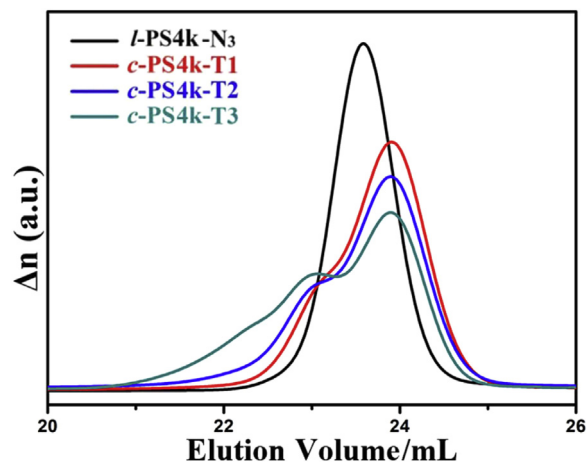


Fig. 5. SEC curves of *l*-PS4k- N_3 precursor and the corresponding cyclization product in toluene at concentrations of 7.9 $\mu\text{mol/L}$ (*c*-PS4k-T1), 15.7 $\mu\text{mol/L}$ (*c*-PS4k-T2), and 31.4 $\mu\text{mol/L}$ (*c*-PS4k-T3).

Unlike the reactions carried out in DMF, all curves in Fig. 5 have shoulder peaks on the left side, while the main peaks from the monocyclic products are overlapped with the curve of corresponding monocyclic PS (*c*-PS-D1). The shoulder peaks became larger with increasing concentration, and were bigger than that in the curves of the corresponding product conducted in DMF, indicating the yield of monocyclic PS products is smaller than in DMF.

Quite astonishingly, for PS precursors with different molecular weights, the one with the lowest molecular weights, i.e., *l*-PS4k- N_3 , showed lower yields than *l*-PS9k- N_3 , where *l*-PS9k- N_3 provided the highest yield of monocyclic PS in toluene. The phenomena can be explained by the “wormlike” chain configuration of PS in solution [63,64]. When the PS chains are too short, the chains behave like semi-rigid chains than the expected flexible chains. Thus the chance of intra-chain head-to-tail “biting” is smaller. This assumption is supported by the measured root-mean-square end-to-end distances $\langle S^2 \rangle^{1/2}$ of linear PS as a function of molecular weights reported by Abe et al. [65], where the data are plotted in Fig. S13. A change of slope is observed at a critical value with the molecular weight around 4000 Da. For PS with higher molecular weights, the slope is 0.59 in good solvent and 0.51 in theta solvent, following the order for traditional flexible chains [4,66]. For those with the molecular weights lower than the critical value, the slope

is 0.89, much greater than the theoretical value for flexible chains, indicating the chains are semi-rigid. In another word, the chain stiffness increases with decreasing molecular weights. Since the chance for intra-chain coupling would be 0 in the case of rigid chains regardless of the end-to-end distance, the increasing chain rigidity will make the chance for the head-to-tail “biting” smaller (note Eq. (1) is only applicable for flexible random coil). This is why *l*-PS9k- N_3 provided the best results among the three PS precursors.

When cyclohexane, a less swelling solvent for PS than toluene and DMF, was used as the reaction solvent for the cyclization of *l*-PS- N_3 , the yield of monocyclic PS is even lower in general than in DMF or toluene. The SEC curves of the corresponding products obtained from *l*-PS4k- N_3 are presented in Fig. 6, while Fig. S14 for products from *l*-PS9k- N_3 and Fig. S15 for *l*-PS14k- N_3 , respectively. The results are summarized in Table 1. The main peak is from the monocyclic products while the shoulder peaks at lower elution volume from the inter-chain coupling products. The yield of monocyclic PS decreased with increasing concentration as predicted. Similar to the reaction carried in toluene, the yield of monocyclic PS synthesized from *l*-PS9k- N_3 is the highest among the three linear PS precursors.

Eq. (3) indicates the yield of monocyclic polymer is only affected by the molar concentration of polymer, and the mean-square end-to-end distance of the linear precursor. The effect of concentration is supported by our experimental results, where the yield of monocyclic polymer decreased with increasing concentration of precursors. On the other hand, since the mean-square end-to-end distance increases with increasing molecular weights for a given polymer, the yield of monocyclic polymer should decrease with increasing molecular weights of the linear precursors. However, such trend was not observed in our system, where the PS precursor with molecular weights of about 9100 Da provided the best results in all the three studied solvents. The reason for the deviation may lie in the “semi-rigid” chain configuration for low molecular weights polymers as discussed previously.

Our experimental results show that the solvents also have great effect on the yield of monocyclic polymers, which has been mentioned by Jia and Monteiro [67] but has not been systematically studied previously. As we know, this is the first time to systematically study the solvent effect on the cyclization yield. However, this effect can hardly be explained by Eq. (3). For PS chains in cyclohexane, they should have the smallest mean-square end-to-end distances than in DMF and toluene, since cyclohexane is a poorer solvent for

Table 1

The yield of monocyclic polystyrenes from corresponding *l*-PS- N_3 linear precursors with different molecular weights in various solvents and concentrations.

Sample ^a	Yield ^b %	Sample ^a	Yield ^b %	Sample ^a	Yield ^b %
<i>c</i> -PS4k-D1	>99	<i>c</i> -PS9k-D1	>99	<i>c</i> -PS14k-D1	99
<i>c</i> -PS4k-D2	98	<i>c</i> -PS9k-D2	98	<i>c</i> -PS14k-D2	95
<i>c</i> -PS4k-D3	96	<i>c</i> -PS9k-D3	97	<i>c</i> -PS14k-D3	94
<i>c</i> -PS4k-T1	79	<i>c</i> -PS9k-T1	93	<i>c</i> -PS14k-T1	90
<i>c</i> -PS4k-T2	63	<i>c</i> -PS9k-T2	90	<i>c</i> -PS14k-T2	76
<i>c</i> -PS4k-T3	55	<i>c</i> -PS9k-T3	85	<i>c</i> -PS14k-T3	72
<i>c</i> -PS4k-C1	70	<i>c</i> -PS9k-C1	91	<i>c</i> -PS14k-C1	64
<i>c</i> -PS4k-C2	62	<i>c</i> -PS9k-C2	82	<i>c</i> -PS14k-C2	49
<i>c</i> -PS4k-C3	61	<i>c</i> -PS9k-C3	74	<i>c</i> -PS14k-C3	46

^a Sample synthesized from corresponding linear PS precursors (*l*-PS- N_3) with molecular weights of 4k, 9k and 14k Da. The suffix of D, T and C means cyclization carried in DMF, toluene and cyclohexane, while number 1, 2 and 3 after the letter means the cyclization concentration of 7.9, 15.7 and 31.4 $\mu\text{mol/L}$, respectively.

^b The area percentage of monocyclic PS calculated from SEC curves by Origin's Fit Multi-Peaks software analysis.

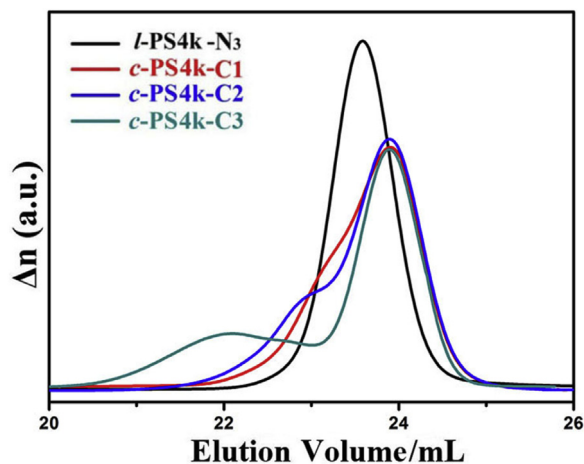


Fig. 6. SEC curves of *l*-PS4k- N_3 precursor and the corresponding cyclization product in cyclohexane at concentrations of 7.9 $\mu\text{mol/L}$ (*c*-PS4k-C1), 15.7 $\mu\text{mol/L}$ (*c*-PS4k-C2), and 31.4 $\mu\text{mol/L}$ (*c*-PS4k-C3).

PS. If we apply this in Eq. (3), the yield of monocyclic PS should be the highest in cyclohexane, which is contrary to the experimental results. One possible reason for it is the chain mobility, which decreases a lot in cyclohexane due to chain contraction. Thus the chance for the intra-chain coupling reduces, while that for the inter-chain coupling is not affected much (the inter-chain coupling reaction is affected by the concentration as predicted by Eq. (2)). Another possible reason is the solubility of Cu(I) salts, which is larger in polar solvent and have a large influence on the “click” reaction rate [33,68]. Thus the cyclization processed much faster in DMF than in toluene and cyclohexane, and the remained concentration of linear precursor was much lower during our solution dropping process, which provided the best monocyclic PS yields.

Though SEC is a facile and relatively reliable method to characterize cyclic polymers, it does not work well when used to detect the purity of cyclic polymers which are synthesized from linear precursors. Liquid chromatography at critical condition (LCCC) is a very efficient method to characterize the purity of cyclic polymers, especially when they are contaminated with linear precursors [4,29,48,49,69,70]. As pointed by D. Yuya and A. Takano, even when samples are purified by SEC fractionation or ultracentrifuge method, and they provide a symmetric curve in SEC, there are still about 10–25% linear contaminations in cyclics when detected by

LCCC [4,69]. We are concerned about the linear precursor contamination in those *c*-PS-D1 samples where the cyclic purity seems to be higher than 99% by SEC. These samples were characterized by LCCC and the results are presented in Fig. 7. The small peaks (denoted as #1) at the left side of major peaks (denoted as #2) are from linear contaminants, since the value is similar to that from corresponding *l*-PS- N_3 linear precursors. The main peaks are cyclic polymers whose elution time increases with the molecular weight. The area percentage of main peak among the total area (#1 + #2) is used to calculate the purity of cyclics, with the corresponding value of 93%, 98% and 95% for *c*-PS4k-D1, *c*-PS9k-D1, and *c*-PS14k-D1, respectively. Though the yield of cyclic PS is lower than the SEC-determined yield listed in Table 1, which indicates that SEC cannot fully resolve cyclic PS from their precursors, these results agree well with the previous discussion, where *c*-PS9k-D1 contains least linear precursors.

It should be noted this is the least percentage of pure cyclic polymers in these samples, due to the unknown small molecules that have strong UV absorption appeared at peak area #1. This is supported by subjecting the collected #2 solutions (pure cyclics) in SEC, and comparing with the area of the original solution (cyclics + linears). The curves are presented in Fig. S16, with the value of 97%, 98% and 96% compared to original peaks for *c*-PS4k-D1, *c*-PS9k-D1, and *c*-PS14k-D1, respectively, which can be considered roughly the percentage of pure cyclics in the samples. Together with the recent reports from Grayson's group who use the similar synthetic strategy, these are among the highest monocyclic polymer yield reported by intra-chain coupling method before further purification using LCCC method [62].

4. Conclusion

In summary, the effect of concentration, solvent and molecular weight on the synthesis of monocyclic PS (*c*-PS) by CuAAC method using fully α -alkynyl- ω -azido functionalized PS as linear precursors was studied. The results revealed the yield of monocyclic PS decreased with increasing PS precursor concentrations as predicted. Quite astonished, the polymer chain conformation and mobility showed significant effect on the *c*-PS yield, where the polar good solvent DMF gave the best yield, and the PS with medium molecular weights among the three studied samples showed the highest *c*-PS yield. By addressing the three concerns related to the intra-chain coupling of linear precursors, a nearly quantitative synthetic approach for *c*-PS was established with the best yield of >99% by SEC and 98% by LCCC, at concentrations around 7.9 $\mu\text{mol/L}$ or lower in DMF solution. This synthetic method excludes almost the contamination of linear polymers in the cyclic products, and opens out a promising method for mass production of cyclic polymers with high purity. Since this is at the scope of synthetic approach by living/controlled free radical polymerization, it would be a promising area in the near future.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.polymer.2016.08.086>.

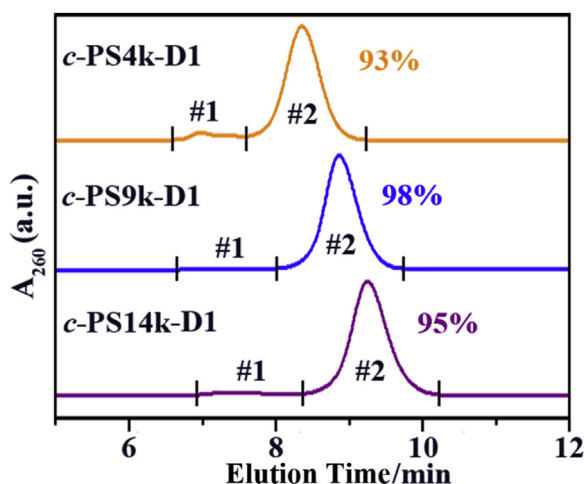


Fig. 7. LCCC curves of *c*-PS-D1 with different molecular weights.

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